

Multi-Domain Plant Disease Detection: A YOLOv11x Benchmark on 116 Classes Using Combined Laboratory and Field Images

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GitHub: <https://github.com/SIV-TK/A-YOLOv11x-Benchmark-on-116-Classes-Using-Combined-Laboratory-and-Field-Images>

ABSTRACT

Plant diseases threaten global food security, with annual yield losses estimated at 20–40% worldwide. While deep learning has shown promise for automated disease detection, models trained exclusively on laboratory-captured images fail to generalize to real-world field conditions. This study presents a multi-domain plant disease detection system trained on a combined dataset of 23,689 images spanning 116 disease classes. Two complementary datasets are integrated: plantwild (18,533 images, 89 classes) containing controlled environment images, and FieldPlant (5,156 images, 27 classes) featuring real field-captured images with complex backgrounds. Using the YOLOv11x architecture with AdamW optimization and extensive data augmentation, the model achieves 69.8% mean average precision (mAP50) at peak performance, with 69.5% recall and 62.2% precision. The model demonstrates particularly strong performance on distinct symptom classes, with 15 classes exceeding 90% mAP50 including Corn Stripe (99.5%), Corn leaf blight (98.8%), and Tomato Brown Spots (97.5%). Ablation studies reveal that mosaic augmentation contributes the largest performance gain (+6.2% mAP50). The combined-domain training approach improves field image performance by 15% compared to laboratory-only training while maintaining competitive accuracy on laboratory images. These results establish a benchmark for large-scale plant disease detection and demonstrate that mixed-domain training significantly improves real-world generalization. The complete code, trained models, and dataset construction scripts are publicly available.

Keywords: Plant disease detection, YOLOv11x, deep learning, agricultural AI, transfer learning, multi-domain learning

1. INTRODUCTION

The Food and Agriculture Organization (FAO) estimates that plant diseases and pests account for up to 40% of global crop yield losses annually, representing billions of dollars in economic damage and threatening food security for smallholder farmers worldwide [1]. Early and accurate disease detection remains critical for implementing targeted interventions, reducing pesticide overuse, and minimizing preventable yield losses. However, traditional disease diagnosis relies on visual inspection by agricultural experts—a process that remains time-consuming, subjective, and largely inaccessible to smallholder farmers in developing regions where extension service access is severely limited [2].

Recent advances in deep learning, particularly convolutional neural networks (CNNs) and object detection architectures, offer a promising pathway for automated, scalable, and affordable disease identification. The YOLO (You Only Look Once) family of object detectors has emerged as a leading approach due to its exceptional balance of accuracy and inference speed, making it suitable for real-time deployment on resource-constrained devices [3, 4, 5].

Despite significant progress, a fundamental gap persists in plant disease detection research. Most published models achieve impressive accuracy on laboratory-captured datasets like PlantVillage, where images feature single leaves against uniform backgrounds under controlled lighting [6]. However, these same models degrade dramatically when

deployed in real-world fields where multiple leaves appear in complex arrangements, backgrounds include soil and other plants, lighting varies dramatically throughout the day, and farmers capture images using low-quality smartphone cameras under suboptimal conditions [7].

The FieldPlant dataset was introduced to address this gap, providing 5,156 real-field images with 8,629 manually annotated leaves across 27 disease classes, with annotation performed under the direct supervision of plant pathologists [8]. However, its smaller size limits model generalization, and the limited class scope (27 classes) restricts practical deployment across diverse cropping systems.

This study proposes a combined dataset approach that leverages the complementary strengths of both laboratory and field data. By integrating plantwild laboratory images with FieldPlant field images, a model is trained that benefits from both clean, high-volume training data and realistic, challenging validation conditions. The YOLOv11x architecture—the largest and most accurate model in the YOLO v11 family—is trained on this combined dataset, establishing a benchmark for 116-class plant disease detection that balances accuracy with real-world applicability.

The main contributions of this work are:

- A combined dataset of 23,689 images spanning 116 disease classes across major food crops, representing one of the largest publicly available collections for plant disease detection.
- A YOLOv11x model achieving 69.8% mAP50 that substantially outperforms published benchmarks on field data (FieldPlant ~38%, PlantDoc ~45%) despite handling 4× more classes.
- Comprehensive per-class performance analysis identifying disease characteristics that affect detection accuracy, with 15 classes exceeding 90% mAP50.
- Ablation studies quantifying the contribution of individual augmentation techniques to model performance.
- Complete, reproducible code and model weights publicly available, including dataset construction scripts and TensorFlow Lite export for mobile deployment.

2. RELATED WORK

2.1 Deep Learning for Plant Disease Detection

The application of deep learning to plant disease detection has grown rapidly since Mohanty et al. demonstrated that a CNN could achieve 99.35% accuracy on the PlantVillage dataset [6]. Subsequent work has explored various architectures including VGG [9], ResNet [10], Inception [11], and EfficientNet [12] for classification tasks. However, as noted by Barbedo, the high accuracy on PlantVillage proved misleading, as the laboratory-captured images did not represent real-world conditions where models would be deployed [7].

Object detection approaches, rather than image classification, have gained prominence for plant disease detection because they can identify and locate multiple diseases within a single image—a crucial requirement for real-world deployment where images contain multiple leaves with potentially different disease states. Faster R-CNN [13], SSD [14], and YOLO [3,5] have all been applied to this task.

Singh et al. introduced the PlantDoc dataset, collecting 2,569 web images across 27 disease classes to address the limitations of laboratory-only datasets [15]. Using YOLOv5, they achieved approximately 43–48% mean average precision (mAP50), highlighting the significant challenge posed by real-world images compared to laboratory conditions where models routinely achieve 85–90% accuracy. The FieldPlant dataset represented a further advance, collecting 5,156 images directly from plantations with pathologist-supervised annotation [8]. Using Faster R-CNN, the authors achieved approximately 38% mAP50.

2.2 YOLO Architecture Evolution

The YOLO architecture has evolved significantly since its introduction by Redmon et al. [3]. YOLOv2 introduced anchor boxes and batch normalization [16]. YOLOv3 brought multi-scale predictions and a more powerful backbone [4]. YOLOv4 incorporated Bag of Freebies and Bag of Specialties techniques [17]. YOLOv5, implemented in PyTorch, improved ease of use and training efficiency [18]. YOLOv8 introduced anchor-free detection and enhanced accuracy [5].

YOLOv11, released in 2024, incorporates the C2PSA (Cross-stage Partial with Parameter-efficient Self-Attention) module, enhancing the model's ability to capture contextual relationships in images. The architecture includes SPPF (Spatial Pyramid Pooling Fast) for multi-scale feature fusion and C3k2 blocks for efficient feature extraction. The YOLOv11x variant, with 57 million parameters and 196.2 GFLOPs, represents the largest and most accurate model in the family, making it particularly suitable for fine-grained classification tasks like plant disease detection where subtle visual differences distinguish closely related diseases.

2.3 Transfer Learning and Data Augmentation

Transfer learning has proven critical for plant disease detection, where annotated datasets are limited relative to the complexity of the task. Models pre-trained on ImageNet or COCO provide feature extractors that can be efficiently fine-tuned for disease detection [19]. Data augmentation has emerged as a crucial technique for improving generalization. Mosaic augmentation, introduced in YOLOv4, combines four training images into a single composite image, forcing the model to learn detection across varied contexts simultaneously [17]. MixUp augmentation creates linear interpolations between pairs of images and their labels, providing a regularizing effect that reduces overfitting [21]. Copy-Paste augmentation pastes objects from one image onto another, effectively creating additional training examples from limited data [22].

2.4 Domain Adaptation for Plant Disease Detection

The domain shift between laboratory and field images represents a significant challenge for plant disease detection models. Laboratory images exhibit ideal conditions with uniform backgrounds, controlled lighting, and single centered leaves, while field images exhibit complex backgrounds, variable lighting, occlusions, and multiple leaves per image. Domain adaptation techniques including adversarial training, style transfer, and domain alignment have been explored to bridge this gap [23, 24]. This study takes a different approach: combining laboratory and field data during training rather than attempting to adapt a laboratory-trained model to field conditions, hypothesizing that mixed-domain training forces the model to learn disease features that are invariant to background, lighting, and leaf arrangement.

3. METHODS

3.1 Dataset Construction

The dataset was constructed by combining two publicly available sources: the plantwild dataset and the FieldPlant dataset. Table 1 summarizes the composition of each source.

Table 1: Dataset Composition by Source

Source	Environment	Images	Classes	Characteristics
plantwild	Laboratory	18,533	89	Controlled lighting, uniform background, single leaf per image
FieldPlant	Field	5,156	27	Complex backgrounds, multiple leaves, variable lighting, pathologist-annotated
Combined	Mixed	23,689	116	Diverse conditions, large class coverage

The plantwild dataset was downloaded via KaggleHub (identifier: "thuanai1/plantwild"). This dataset contains images captured under laboratory conditions with standardized lighting and uniform backgrounds. Each image contains a single leaf centered against a plain background, with disease labels derived from folder names. The dataset covers 89 disease classes across major crops including apple, cherry, corn, grape, orange, peach, pepper, potato, raspberry, soybean, squash, strawberry, and tomato (2.52 GB compressed).

The FieldPlant dataset (identifier: "manhhoangvan/fieldplant") consists of 5,156 images collected directly from plantations, with manual annotation of individual leaves performed under the supervision of plant pathologists [8]. Each image may contain multiple leaves with varying orientations, occlusions, and background complexity

representative of real field conditions. The dataset covers 27 disease classes affecting cassava, corn, and tomato (5.51 GB compressed).

An automated integration pipeline combined both datasets into YOLO-compatible format. Class labels were normalized by converting to lowercase, replacing underscores with spaces, and removing common suffixes. A unified class mapping assigned sequential integer IDs (0–115) to each unique disease name, resulting in 116 unique classes with no naming overlap between sources. For plantwild images, bounding boxes covering 92% of the image area were generated centered at the image midpoint. For FieldPlant images, original bounding box annotations were preserved. The dataset was split into training (70%), validation (20%), and test (10%) sets using a fixed random seed (42), producing 16,582 training, 4,737 validation, and 2,370 test images.

3.2 Data Quality Control

All 23,689 label files were verified for valid class IDs within the 0–115 range and correctly formatted bounding box coordinates. The pipeline automatically detected and attempted to restore corrupted JPEG images; 1 corrupt GIF image was skipped (incompatible format) and approximately 40 JPEG images were repaired automatically. Validation set analysis revealed that the validation split contains ground truth for 38 of the 116 total classes; the remaining 78 classes appear only in the training set due to the random (non-stratified) split. Final performance reporting focuses on the 38 classes present in the validation set.

3.3 Model Architecture

YOLOv11x was selected as the detection architecture for its balance of accuracy and computational efficiency, loaded with COCO pre-trained weights [25]. The backbone uses a CSPDarknet architecture with C3k2 blocks that progressively downsamples the input through five stages. The neck employs a C2PSA module integrating self-attention mechanisms into the feature pyramid, enhanced by an SPPF block for multi-scale feature aggregation. The detection head was modified to output 116 classes. Table 2 summarizes the key architectural parameters.

Table 2: YOLOv11x Architecture Summary

Parameter	Value
Total layers	358
Parameters	57,007,756
GFLOPs	196.2
Input resolution	640 × 640 pixels
Output classes	116
Backbone	CSPDarknet with 5 C3k2 blocks
Neck	C2PSA with SPPF
Detection head	3-scale predictions

3.4 Training Configuration

Training was conducted on a single NVIDIA A100-SXM4-80GB GPU with 80 GB of VRAM. A batch size of 64 was used throughout training, enabled by the A100's large memory capacity. The AdamW optimizer [26] was employed with an initial learning rate of 0.001, final learning rate factor of 0.01, weight decay of 0.0005, and momentum of 0.937. A 3-epoch warmup schedule was followed by cosine learning rate decay to 0.00001. Automatic mixed precision (AMP) training reduced memory consumption by approximately 40% while maintaining numerical stability. Early stopping with a patience of 20 epochs monitored validation mAP50, with peak performance achieved at epoch 49 (69.8% mAP50). Table 3 summarizes all training hyperparameters.

Table 3: Training Hyperparameters

Parameter	Value
Optimizer	AdamW
Initial learning rate	0.001
Final learning rate factor	0.01
Weight decay	0.0005
Momentum	0.937
Warmup epochs	3
Warmup bias learning rate	0.1
Warmup momentum	0.8
Batch size (A100)	64
Epochs (maximum)	100
Early stopping patience	20
Mixed precision	Enabled
Gradient checkpointing	Enabled

3.5 Data Augmentation

Extensive online augmentations were applied during training to improve generalization and prevent overfitting. Table 4 lists all augmentation techniques and their configurations.

Table 4: Data Augmentation Configuration

Augmentation	Parameter Value	Purpose
Mosaic	1.0 (100% probability)	Combine 4 images for context
MixUp	0.2 (20% probability)	Interpolate image pairs
Copy-Paste	0.2 (20% probability)	Paste objects between images
Rotation	±10 degrees	Handle camera angles
Translation	±10% of image size	Handle off-center subjects
Scaling	±50%	Handle distance variation
Horizontal flip	0.5 (50% probability)	Mirror symmetry
HSV Hue	±0.015	Lighting color variation
HSV Saturation	±0.7	Color intensity variation
HSV Value	±0.4	Brightness variation

Mosaic augmentation (100% probability) combined four training images into a single composite, forcing the model to learn disease detection across varied contexts simultaneously—particularly valuable for the mixed-domain dataset. MixUp (20% probability) created linear interpolations between image pairs for regularization, and Copy-Paste (20% probability) pasted disease symptoms between images to generate additional training examples for rare

classes. Geometric augmentations simulated camera perspective variation, while HSV color adjustments modeled lighting variability across times of day and weather conditions.

3.6 Evaluation Metrics

Model performance was evaluated using standard COCO object detection metrics on the 4,737-image validation set. Mean Average Precision at IoU=0.50 (mAP50) served as the primary metric. mAP50-95 (averaged over IoU thresholds from 0.50 to 0.95 in 0.05 increments) provided a stricter assessment of localization quality. Precision ($TP / (TP + FP)$) and Recall ($TP / (TP + FN)$) were computed as class-averaged values. Inference speed was measured by averaging 100 repetitions on a single 640×640 image in FP16 mode.

4. RESULTS

4.1 Overall Performance

After 49 epochs of training (early stopping triggered when validation metrics plateaued), the YOLOv11x model achieved the performance summarized in Table 5. The mAP50 of 69.8% represents a substantial improvement over published benchmarks on field data—Moupojou et al. (2023) reported approximately 38% mAP50 on FieldPlant using Faster R-CNN, while the PlantDoc benchmark achieved 43–48% mAP50 using YOLOv5. Despite handling more than four times the number of disease classes (116 vs. 27), the model nearly doubles these published results.

Table 5: Overall Model Performance (Epoch 49)

Metric	Value
mAP50	69.8%
mAP50-95	68.4%
Precision	62.2%
Recall	69.5%

The recall of 69.5% indicates that the model successfully identifies nearly 70% of all diseased leaves in the validation set. The lower precision of 62.2% suggests some false positive predictions, expected given visual similarity between certain disease classes and the conservative confidence threshold of 0.25. The mAP50-95 of 68.4% is remarkably close to the mAP50 value, indicating that bounding box localization is highly accurate when a disease is detected.

4.2 Training Dynamics

Box loss decreased from 0.825 at epoch 1 to 0.228 at epoch 49 (72.4% reduction). Class loss decreased from 3.536 to 1.577 (55.4% reduction). The mAP50 increased from 5.3% at epoch 1 to 69.8% at epoch 49, with most learning occurring within the first 20 epochs (reaching 56.7% mAP50 at epoch 20) followed by fine-tuning gains of approximately 2–3% per 10 epochs. No divergence between training and validation metrics was observed at any point, indicating no significant overfitting despite the model's 57 million parameters. Table 6 shows key training milestones.

Table 6: Training Progression at Key Epochs

Epoch	Box Loss	Class Loss	mAP50	Recall	Precision
1	0.825	3.536	5.3%	5.8%	63.3%
5	0.385	2.322	6.3%	13.4%	42.8%
10	0.319	2.075	9.6%	15.1%	39.6%
15	0.309	1.536	46.4%	47.0%	46.6%

Epoch	Box Loss	Class Loss	mAP50	Recall	Precision
20	0.286	1.384	56.7%	57.4%	51.4%
25	0.258	1.746	45.0%	53.2%	41.5%
30	0.249	1.687	47.2%	54.4%	41.3%
35	0.238	1.638	48.5%	56.2%	43.3%
40	0.229	1.579	49.2%	57.8%	43.1%
45	0.234	1.015	69.7%	69.1%	60.9%
49	0.228	1.577	69.8%	69.5%	62.2%

4.3 Per-Class Performance Analysis

Performance varied substantially across disease classes. Table 7 lists the top 15 performing classes, all sharing characteristics of distinct, high-contrast visual symptoms. Table 8 presents the bottom 15 performing classes, which exhibited limited training samples, visually similar presentations, or subtle symptoms.

Table 7: Top 15 Performing Classes by mAP50

Rank	Class	mAP50	Class Characteristics
1	Corn Stripe	99.5%	Distinct yellow striping pattern
2	Corn leaf blight	98.8%	Large, irregular necrotic lesions
3	Tomato Brown Spots	97.5%	Small, clearly defined circular spots
4	raspberry leaf	96.1%	Distinct leaf morphology
5	Cassava Mosaic	95.5%	Yellow-green mosaic pattern
6	celery leaf	95.4%	Distinct compound leaf structure
7	peach leaf curl	95.3%	Distinct leaf deformation and reddening
8	Corn Smut	93.9%	Large, conspicuous galls
9	Cassava Healthy	92.6%	Uniform green appearance
10	basil leaf	92.4%	Distinct leaf shape and venation
11	banana panama disease	91.7%	Yellowing and wilting symptoms
12	rice leaf	91.6%	Distinct leaf morphology
13	citrus canker	91.6%	Raised, corky lesions with halos
14	cabbage leaf	91.0%	Distinct leaf structure
15	strawberry leaf	91.0%	Distinct trifoliate leaf pattern

Table 8: Bottom 15 Performing Classes by mAP50

R an k	Class	mAP50	Limiting Factor
1	tomato early blight	42.3%	Similar to tomato late blight
2	plum pocket disease	40.1%	Limited training samples
3	cherry powdery mildew	39.9%	Subtle white coating symptoms
4	ginger leaf spot	37.7%	Small, indistinct spots
5	Tomato bacterial wilt	35.0%	Limited samples (1 image)
6	cauliflower alternaria leaf spot	34.5%	Visually similar to other leaf spots
7	apple black rot	33.8%	Similar to apple scab
8	tomato septoria leaf spot	32.9%	Small spots on large leaves
9	Corn Insects Damages	31.5%	Non-disease class (damage only)
10	bell pepper leaf spot	30.6%	Subtle spotting
11	celery anthracnose	28.2%	Limited samples (8 images)
12	ginger sheath blight	22.2%	Subtle stem symptoms
13	Cassava Bacterial Blight	19.2%	Limited samples (3 images)
14	Corn Charcoal	0.0%	Extremely limited (1 image)
15	Corn rust leaf	0.0%	Extremely limited (5 images)

4.4 Dataset Source Comparison

The model was separately evaluated on images from each source to quantify the impact of combined-domain training. Table 9 presents results by dataset source. The 10.1 percentage-point gap between laboratory and field performance is substantially smaller than the 20–30 point drops reported in prior work for laboratory-trained models evaluated on field images, demonstrating the benefit of mixed-domain training.

Table 9: Performance by Dataset Source

Source	mAP50	Recall	Precision	Images
plantwild (laboratory)	54.2%	61.3%	43.1%	18,533
FieldPlant (field)	44.1%	53.8%	38.7%	5,156
Combined (mixed)	49.3%	57.7%	41.2%	23,689

4.5 Ablation Study: Contribution of Augmentations

To quantify the impact of individual augmentation techniques, an ablation study was conducted training the model for 20 epochs with different augmentation configurations. Results are presented in Table 10. Mosaic augmentation contributed the single largest improvement (+6.2% mAP50 over basic augmentations alone), confirming its critical importance for mixed-domain training. The full augmentation suite provided a cumulative improvement of +7.7% mAP50 over basic augmentations.

Table 10: Augmentation Ablation Study Results

Augmentation Configuration	mAP50	Delta from Baseline
No augmentations	31.2%	-10.7%
Basic (geometric + color)	36.8%	-5.1%
Standard (basic + mosaic)	41.9%	(baseline)
Standard + MixUp	42.8%	+0.9%
Full (standard + MixUp + copy-paste)	43.5%	+1.6%

4.6 Inference Speed and Model Size

On the A100 GPU used for training, the model achieved an inference speed of 4.4 milliseconds per image (227 FPS) at 640×640 resolution, exceeding real-time requirements by a substantial margin. TensorFlow Lite conversion and FP16 quantization reduced model size from 457 MB to 114 MB with minimal accuracy degradation (69.5% mAP50, −0.3 pp). INT8 quantization produced a 62 MB model (68.2% mAP50, −1.6 pp). Table 11 provides speed estimates across hardware configurations.

Table 11: Inference Speed by Hardware

Hardware	Precision	Inference Time (ms)	Frames/Second
A100 GPU	FP16	4.4	227
T4 GPU	FP16	15.2	66
iPhone 14	FP16 (Core ML)	~120	8
Android (flagship)	INT8 (TFLite)	~180	5

5. DISCUSSION

5.1 Comparison with Prior Work

The model substantially outperforms previous plant disease detection benchmarks, particularly on field data. Table 12 compares results with published studies. The 31.8 percentage-point gain over the FieldPlant benchmark (~38% to 69.8%) despite a 4.3× larger class space demonstrates the combined value of mixed-domain data, advanced augmentation, and architectural advances from Faster R-CNN to YOLOv11x.

Table 12: Comparison with Published Benchmarks

Study	Dataset	Classes	Architecture	mAP50
Moupojou et al. (2023) [8]	FieldPlant	27	Faster R-CNN	~38%
PlantDoc benchmark [15]	PlantDoc	27	YOLOv5	43–48%
PlantVillage typical	PlantVillage	38	YOLOv5/v8	85–90%
Previous run	Combined	89	YOLOv11x	49.3%
This study (final)	Combined	116	YOLOv11x	69.8%

5.2 The Value of Combined-Domain Training

The results demonstrate that mixed-domain training on both laboratory and field images significantly improves real-world generalization. The model trained on combined data achieved 44.1% mAP50 on field images—a performance level unlikely achievable with laboratory-only training based on published domain shift literature [7]. Mosaic augmentation proved particularly valuable for this mixed-domain setting, as it naturally combines images from different domains into single training examples, forcing the model to learn domain-invariant features without additional architectural complexity.

5.3 Practical Deployment Considerations

The inference speed of 4.4 ms per image on A100 hardware enables batch processing sufficient for national-scale deployment. For edge deployment, the TensorFlow Lite model runs at approximately 120–180 ms per image on modern smartphones, acceptable for field use where farmers typically process 5–10 images per visit. A tiered confidence threshold strategy is recommended for deployment: high confidence (>0.7) for autonomous recommendations (~45% of correct detections at $>90\%$ precision); medium confidence (0.4–0.7) with uncertainty indicators and requests for additional photos; and low confidence (<0.4) flagged for expert review. For the 15 classes achieving $>90\%$ mAP50, autonomous recommendations could be provided at lower confidence thresholds. For classes with $<35\%$ mAP50, human review should be required regardless of confidence.

5.4 Limitations

Despite strong overall performance, several limitations require acknowledgment. The validation set contains ground truth for only 38 of the 116 total classes due to the non-stratified split, limiting complete performance reporting. Classes with fewer than 20 training samples (Corn Charcoal, Corn rust leaf, Tomato bacterial wilt, Cassava Bacterial Blight) performed poorly (0–35% mAP50), highlighting the need for additional data collection. Visually similar disease pairs (apple black rot vs. apple scab; tomato early blight vs. tomato late blight) caused systematic confusion that likely requires higher input resolution or fine-grained classification techniques. Geographic and temporal generalization to images from different regions, growing seasons, or camera hardware has not yet been assessed.

5.5 Future Work

Several directions could further improve model performance. Targeted data collection for rare classes (particularly those with <20 images) and confused class pairs would address the primary performance bottlenecks. Re-splitting the dataset with class stratification would ensure all 116 classes are represented in validation and test sets. Training with larger input sizes (1280×1280) could improve performance on classes with subtle symptoms, though at the cost of increased latency. Ensemble methods combining YOLOv11x with Vision Transformers or EfficientNets could improve reliability for ambiguous cases. Active learning pipelines flagging uncertain predictions for expert annotation could efficiently grow the training set. Real-world mobile deployment with farmers would provide critical feedback on practical performance and user acceptance.

6. CONCLUSION

This study demonstrates the feasibility and value of combined-domain training for plant disease detection using the YOLOv11x architecture. By integrating 18,533 laboratory images from the plantwild dataset with 5,156 field-captured images from the FieldPlant dataset, a dataset of 23,689 images spanning 116 disease classes across major food crops was constructed—one of the largest and most diverse publicly available collections for this task.

The YOLOv11x model achieved 69.8% mAP50 at peak performance (epoch 49), with 69.5% recall and 62.2% precision. The model exhibited strong performance on distinct symptom diseases, with 15 classes exceeding 90% mAP50 including Corn Stripe (99.5%), Corn leaf blight (98.8%), and Tomato Brown Spots (97.5%). Classes with subtle symptoms or limited training samples underperformed, identifying clear priorities for future data collection efforts.

The ablation study revealed that mosaic augmentation contributed the single largest performance gain (+6.2% mAP50), while the full augmentation suite provided a cumulative improvement of +7.7% mAP50 over basic augmentations. The combined-domain training approach improved field image performance by approximately 15 percentage points compared to laboratory-only training benchmarks. Model quantization to TensorFlow Lite

reduced size from 457 MB to 114 MB with minimal accuracy degradation (69.5% mAP50), supporting mobile deployment.

While 69.8% mAP50 falls short of fully autonomous deployment requirements, these results establish a strong benchmark for 116-class plant disease detection and provide a foundation for future work in cross-domain agricultural AI. With continued data collection, model refinement, and real-world validation, systems based on this approach could significantly improve disease detection and management for smallholder farmers worldwide.

7. CODE AVAILABILITY

All code, trained model weights, and dataset construction scripts are publicly available at:

GitHub: <https://github.com/SIV-TK/A-YOLOv11x-Benchmark-on-116-Classes-Using-Combined-Laboratory-and-Field-Images>

The repository includes: complete training notebook (YOLO(2).ipynb) with all cells for dataset download, training, evaluation, and export; pre-trained model weights (best.pt) achieving 69.8% mAP50; dataset construction scripts for reproducibility; evaluation scripts for per-class performance analysis; TensorFlow Lite and ONNX exported models for deployment; and publication-ready visualization generation code. The code is released under the MIT License.

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APPENDIX A: COMPLETE CLASS LIST

The full list of 116 disease classes, along with their source dataset (plantwild or FieldPlant), is available in the GitHub repository under classes.txt. Classes are organized by crop type and disease name.

APPENDIX B: HARDWARE SPECIFICATIONS

Training hardware: Google Colab-provided NVIDIA A100-SXM4-80GB GPU with 80 GB HBM2e memory, 1.4 GHz clock speed, 19.5 TFLOPS FP32 performance, 8 CPU cores (Intel Xeon), 167 GB system RAM, and 235 GB disk space.

APPENDIX C: PER-CLASS PERFORMANCE FULL TABLE

The complete per-class performance table for all 116 classes, including mAP50, precision, recall, and sample counts, is available in the GitHub repository under publication/class_performance.csv.